

Nicola Conci

Full Professor candidate, Associate Professor

Department of Information Engineering and Computer Science (DISI), University of Trento

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150+ publications	10+ PhD students	150+ theses supervised	20+ years of research
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Profile

Nicola Conci is Associate Professor at the University of Trento, where he works in the Multimedia Signal Processing and Understanding Lab (MM-Lab). His research focuses on Computer Vision, Artificial Intelligence, Human Behaviour Analysis, Digital Twins, Crowd Understanding, 3D Reconstruction, Extended Reality and Multimedia Signal Processing. He has authored more than 150 scientific publications and has coordinated and participated in several national and European research projects.

Research Areas

Computer Vision; Artificial Intelligence; Human Behaviour Understanding; Crowd Analysis and Simulation; Digital Twins; 3D Reconstruction and Neural Rendering; Extended Reality (AR/VR/XR); Multimedia Signal Processing; Sports Analytics; Assistive Technologies.

Career Timeline

- 2018-present - Associate Professor, University of Trento.
- 2009-2018 - Assistant Professor.
- 2008-2009 - Postdoctoral Researcher, Queen Mary University of London.
- PhD, University of Trento.

Selected Publications

- Borges, Conci, Cavallaro. "Video-Based Human Behavior Understanding: A Survey." IEEE Transactions on Circuits and Systems for Video Technology, 2013.
- Sansone, Apostolidis, Conci et al. "Automatic Synchronization of Multi-user Photo Galleries." IEEE Transactions on Multimedia, 2017.
- Bisagno, Saltori, Zhang, De Natale, Conci. "Embedding Group and Obstacle Information in LSTM Networks for Human Trajectory Prediction in Crowded Scenes." Computer Vision and Image Understanding, 2021.
- Zhang, Wang, Zhou, Conci, Liu. "Human Trajectory Forecasting Using a Flow-Based Generative Model." Engineering Applications of Artificial Intelligence, 2022.
- Garau, Conci. "CapsulePose: A Variational CapsNet for Real-Time End-to-End 3D Human Pose Estimation." Neurocomputing, 2023.
- Orlandi, Depedri, Conci. "Spatial-Temporal Calibration for Outdoor Location-Based Augmented Reality." IEEE Sensors Journal, 2024.
- Martinelli, Garau, Bisagno, Conci. "MoMa: Skinned Motion Retargeting Using Masked Pose Modeling." Computer Vision and Image Understanding, 2024.
- Sambugaro, Garau, Bisagno, Conci. "Agglomerator++: Interpretable Part-Whole Hierarchies and Latent Space Representations in Neural Networks." Computer Vision and Image Understanding, 2024.

Selected Projects

- TWINMAN (2024-2025): Human Digital Twins and behavioural modelling within the RESTART program.
- SEQUOIA (PRIN 2022): Semantic and Quality-Oriented View Reconstruction Algorithms for neural rendering and 3D scene reconstruction.
- DIMOTY: AI-based outdoor 3D reconstruction using Neural Radiance Fields and digital twins.

- ANTEMOTION: Simulation environments for autonomous driving, traffic modelling and behavioural simulation.
- UNCAP (Horizon 2020): Technologies for active ageing and independent living.
- Citizen Safety (EIT): Crowd monitoring, risk assessment and emergency management in large-scale events.

Teaching and Service

- Current courses: Computer Vision; Vision and Recognition; Signal Processing; Sport Tech.
- Supervisor of PhD students.
- Supervisor of more than 150 MSc and BSc theses.
- Executive Committee member of the ICT Doctoral School.
- General Chair, ICDSC 2019.
- Area Chair, ICIP and ICASSP.
- Co-Chair, Workshop on Computer Vision for Winter Sports (WACV).
- Tutorial speaker at ICIP and ACM Multimedia Asia.
- Associate Editor, Signal Processing: Image Communication.

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